



GlusterFS: A Dependable Distributed File System

The Gluster file system is an open source distributed file, which can scale out in a 'building block' manner to store several petabytes of data. This article will be of interest to newbies and those who wish to know more about file systems.



A distributed file system gives us a way of storing and accessing files in a client-server architecture. We can use more than one server to store data and use multiple clients (local or remote) that can access data from these servers. A distributed file system organises and displays files and directories from multiple servers as if they were stored in your local system, thereby projecting a simple interface to the user or application. The main advantage is that it is easier to distribute documents to multiple clients and provide a centralised storage system so that client machines do not use their resources to store the data.

GlusterFS concepts

Some of the basic terminology that we need to know to understand GlusterFS is given in Table 1.

Introducing GlusterFS

GlusterFS is a distributed file system that can scale up to several petabytes and can handle thousands of clients. It

clusters together storage building blocks over RDMA or TCP/IP, and aggregates disk and memory resources in order to manage data in a single global namespace. These blocks do not need any special hardware to store data and are based on a stackable user space design. GlusterFS is a user space file system and hence uses FUSE (file system in user space) to hook itself with the virtual file system (VFS) layer. FUSE is a loadable kernel module that lets non-privileged users create their own file systems without editing kernel code. Gluster uses already tried and tested disk file systems like ext3, ext4, xfs, etc, as the underlying file system to store the data.

Daemons

Before we can create a volume, we must have a trusted storage pool consisting of storage servers that provide bricks to create the volume. Every node has a daemon - *glusterd* - running (this is a management daemon). Each *glusterd* can interact with the *glusterd* on other nodes to add them to its trusted storage pool. This is called peer probing. A node can have multiple